InternSpark Internship

Task 1 — Machine Learning Classification

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Project: ML Classification using Scikit-learn

This report documents the dataset preparation, model training, cross-validation, evaluation, and comparison performed for Task 1.

1. Objective

The objective of this task was to build and evaluate machine learning classification models using preprocessing, train-test splitting, cross-validation, and standard classification metrics.

2. Dataset

Property	Value
Dataset	Breast Cancer Wisconsin Dataset (Scikit-learn)
Total samples	569
Input features	30
Target classes	2
Missing values	0

3. Preprocessing

• The target column was separated from the 30 input features.

• The data was split into 80% training data (455 samples) and 20% testing data (114 samples).

• StandardScaler was fitted on the training data and then applied to both training and testing data.

• The training/testing split used stratification to preserve the target-class distribution.

4. Machine Learning Models

Two classification algorithms were trained and compared: Logistic Regression and Random Forest Classifier.

5. Cross-Validation

A 5-Fold Cross-Validation was performed on the training data for Logistic Regression. The five fold accuracies were 96.70%, 97.80%, 96.70%, 100.00%, and 98.90%. The resulting mean cross-validation accuracy was **98.02%**.

6. Model Evaluation Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	98.25%	98.61%	98.61%	98.61%	99.54%
Random Forest	95.61%	95.89%	97.22%	96.55%	99.39%

Best-performing model: Logistic Regression, with 98.25% test accuracy and 99.54% ROC-AUC.

7. Model Comparison Screenshot

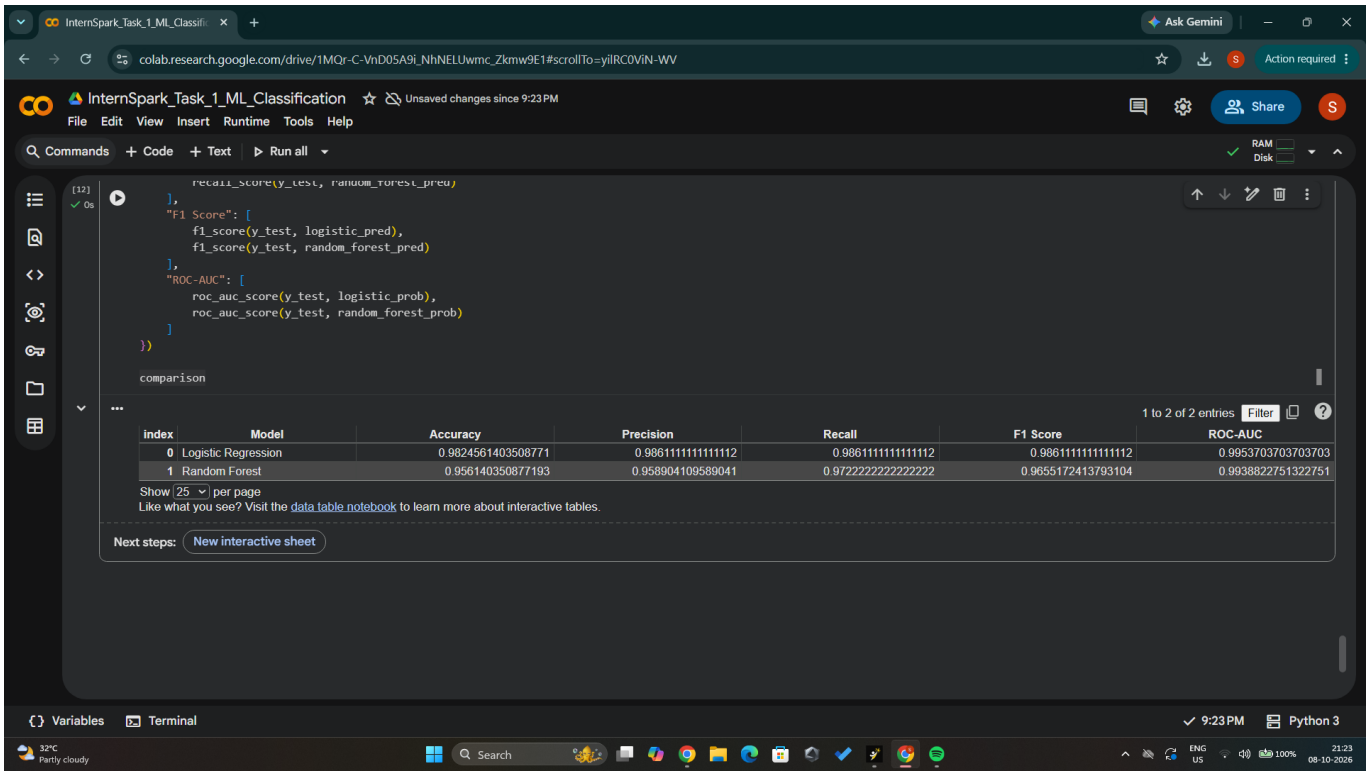


Figure 1. Final comparison table generated in Google Colab.

8. Confusion Matrix

The Logistic Regression confusion matrix shows 41 correct Class 0 predictions, 71 correct Class 1 predictions, and only 1 misclassification in each direction.

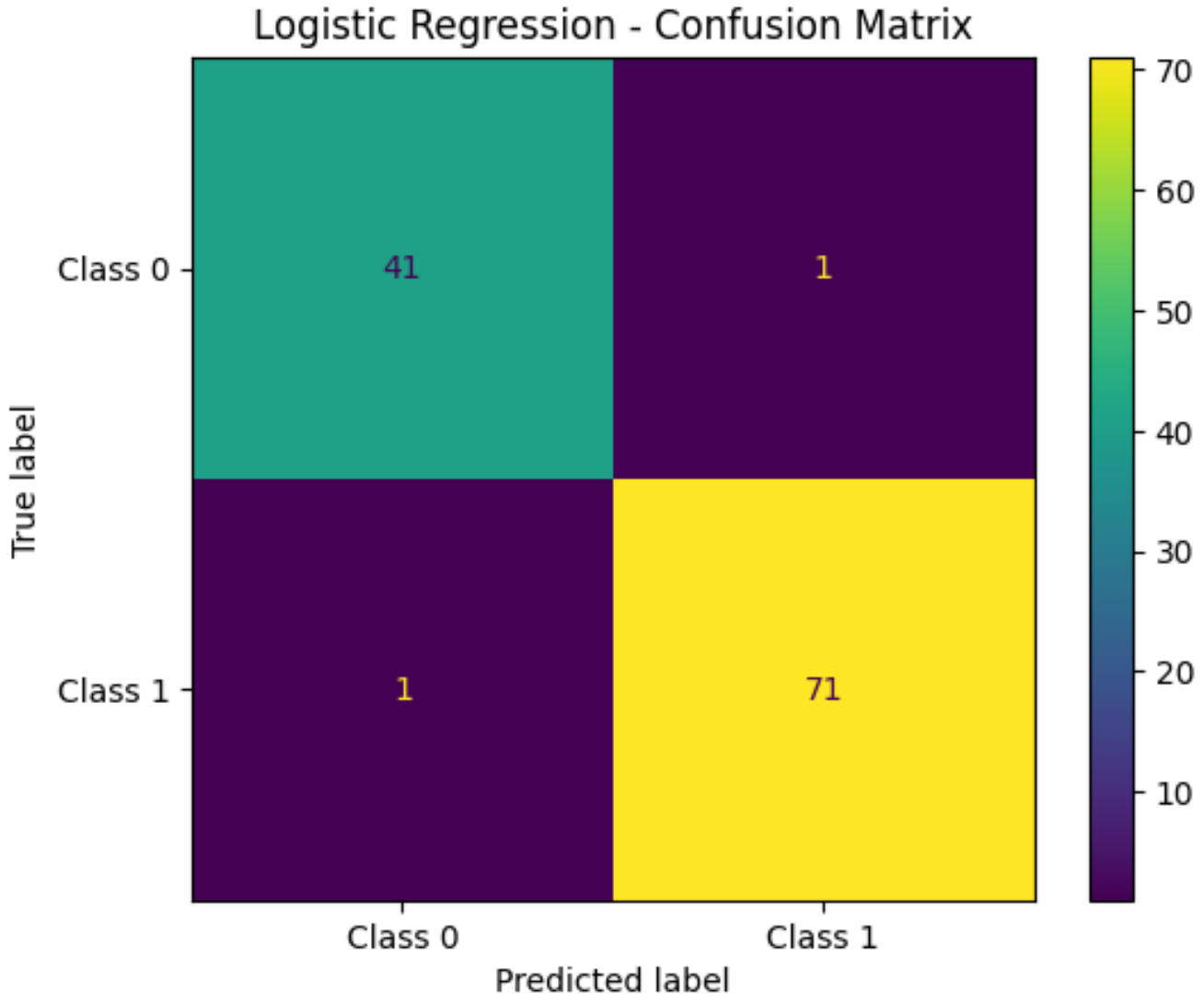


Figure 2. Logistic Regression confusion matrix on the test set.

9. ROC Curve

The ROC curve for Logistic Regression lies close to the upper-left region of the plot, consistent with the measured ROC-AUC of 0.9954 (99.54%).

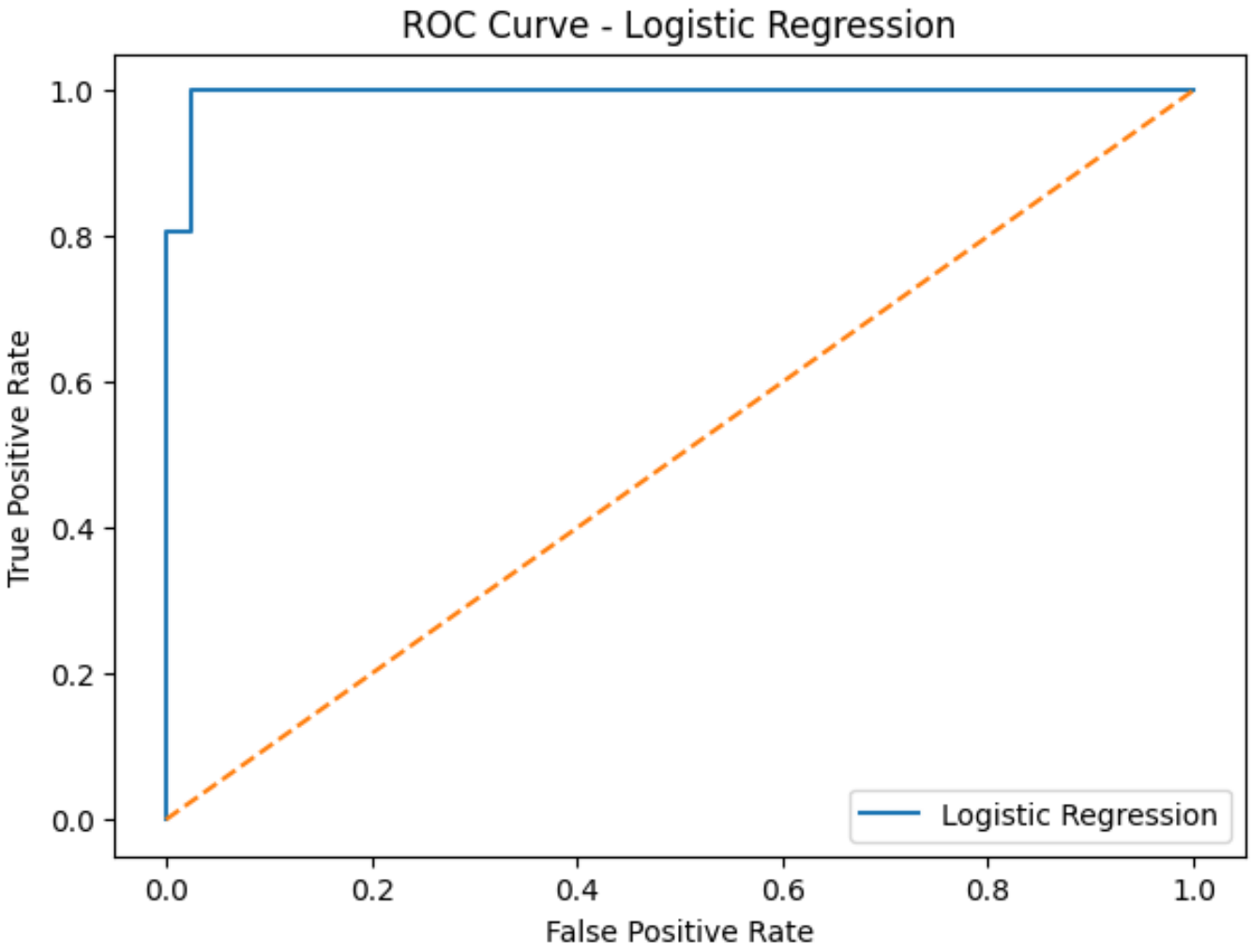


Figure 3. ROC curve for Logistic Regression.

10. Conclusion

Both classification models achieved strong performance on the selected dataset. Logistic Regression performed better than Random Forest across the main reported test metrics, achieving 98.25% accuracy, 98.61% precision, 98.61% recall, 98.61% F1 score, and 99.54% ROC-AUC. Its 5-Fold Cross-Validation mean accuracy was 98.02%. Therefore, Logistic Regression was selected as the best-performing model for this task.

11. Tools & Technologies

Python • Pandas • NumPy • Scikit-learn • Matplotlib • Google Colab

12. Project Files

The project folder contains the Google Colab notebook and README.md. The final report is provided as this PDF document.